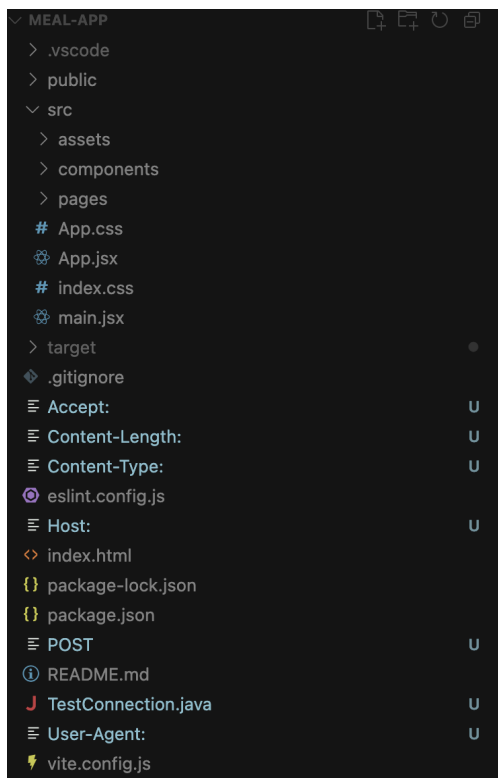


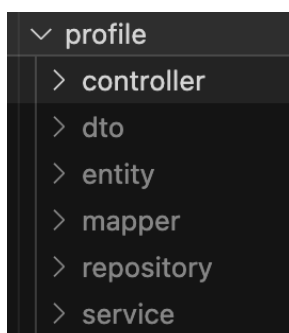
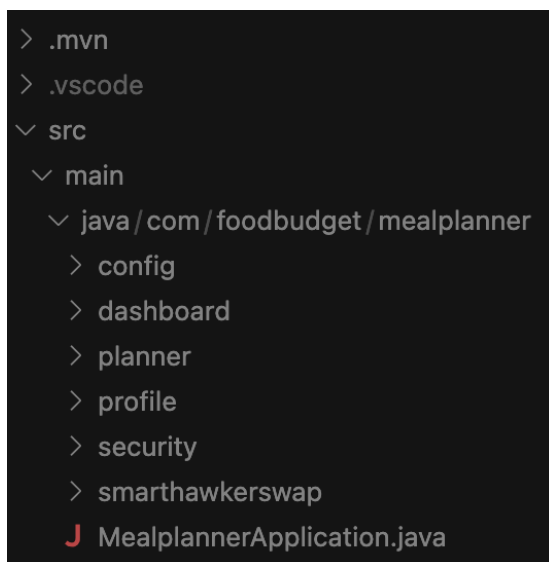
Frontend



The Frontend is a React + Vite single - Page application with a modular, feature - oriented layout. Each feature is implemented as a set of pages and reusable components, wired together using React Router.

- Each major feature of the system such as
 - Profile, BudgetMealPlanner, INflation Dashboard etc has its own page under /pages, which is supported by elements in /components.
 - The Navbar provides consistent navigation.
- Other feature pages include its own form logic, result views and css styling.
- The /assets folder stores shared icons and images, and /App.jsx handles route configuration using the React Router.

Backend



The structure for backend follows a modular, layered design based on the project's five main features. Each feature module follows a similar structure for consistency and maintainability.

- The /profile module manages user profiles, saved meal plans, grocery lists, and preference settings with autosave and encryption.
 - The /planner module handles the Budget Meal Planner, generating meal plans based on user budget, dietary rules, and calorie preferences.
 - The /grocery module powers the Grocery Basket Planner, consolidating ingredients, scaling quantities, and estimating grocery costs.
 - The /dashboard module visualizes CPI-based food price inflation trends and supports filtering and comparison across time ranges.
 - The /smarthawkerswap module enables Smart Hawker Swap recommendations, suggesting cheaper hawker meals within a 3 km radius.
- Lastly, /config contains application-wide configurations, /security ensures user data protection.